Distributed coding of choice, action and engagement across the mouse brain

A figure-by-figure reproduction, with responses in VISp, LGd and LP

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Abstract

This report reproduces the analyses of a brain-wide Neuropixels study of visual decisions in mice [1], from its public dataset [2], and redoes four of its plots for primary visual cortex (VISp) and two visual thalamic nuclei (LGd and LP). All 39 sessions were analysed: 9,538 trials and 29,178 neurons, of which 22,210 were assigned to the study's 42 regions. Each figure was then compared with the original panel by panel. The main results hold: action coding is global, visual coding follows a restricted pathway, choice coding is rare and localized, midbrain neurons prefer contralateral choices, and pre-stimulus engagement signals have opposite signs in neocortex and in subcortical regions. In VISp, LGd and LP, activity before the stimulus is lower in the task than in passive replay, and these regions carry visual and movement signals but very little choice coding. Some numbers differ from the paper, and those differences are listed.

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1 Summary of the comparison

Every panel group of Figures 1 to 5 and Extended Data Figures 1 to 9 was placed beside the original and rated. **Same** means the result cannot be told apart at reading accuracy. **Close** means the same pattern and conclusion, with small differences in values. **Partial** means the main conclusion holds but a difference is visible. **From the paper** marks a photo, histology image or cartoon with no data behind it: these are crops of the original figure, tagged "from the paper" in the figure itself.

Figure	Same	Close	Partial	From the paper
Figure 1	0	4	1	5
Figure 2	0	7	1	0
Figure 3	1	6	0	0
Figure 4	0	7	1	0
Figure 5	0	5	1	0
Extended Data Figure 1	1	5	1	0
Extended Data Figure 2	0	2	0	3
Extended Data Figure 3	0	0	0	1
Extended Data Figure 4	0	3	1	0
Extended Data Figure 5	0	6	1	1
Extended Data Figure 6	0	2	0	0
Extended Data Figure 7	0	6	0	0
Extended Data Figure 8	0	3	2	0
Extended Data Figure 9	0	6	1	0
Panel groups	2	62	10	10

Table 1: Agreement with the paper by figure. Ratings are by eye, supported by the numbers in Section 5.

The ratings for the example neurons need care. Those neurons were chosen to resemble the paper's panels (Section 4), so their close match shows that such neurons exist in the data. It is not independent evidence for the paper's claims. The evidence is in the population panels, where nothing was selected.

The five largest differences from the paper are these:

1. **Fig 4c:** the time-by-group interaction is absent as in the paper ($p = 1.0$), but the group main effect is significant here ($p = 7e\text{-}4$). The paper reports none. So the conclusion that choice signals have the same timing across regions holds, while the claim of no group difference does not fully reproduce.
2. **Choice-selective neurons are fewer:** 79 in the midbrain and forebrain sets against the paper's 102, although the proportions preferring each side are close.
3. **Suppression of the non-preferred choice:** 16 of 46 midbrain neurons against 29 of 54. This is the largest proportional gap in the single-neuron results.
4. **Extended Data Table 1:** 23 of the 42 regions differ by more than 2%, because the released region labels differ from those behind the paper's table.
5. **Fig 5d and Ext 7f:** the correlations are stronger than the paper's ($r = 0.58$ against 0.46, and 0.79 against 0.69).

2 Data and methods

2.1 Dataset

The data are 39 sessions from 10 mice, recorded with Neuropixels probes in the left hemisphere [2]. On each trial, gratings of varying contrast could appear on the left, the right, both or neither, and mice turned a wheel

to bring the higher-contrast grating to the center, or held still if there was none. Because all recordings are from the left hemisphere, *contra* means the right stimulus and the Right choice. The sign conventions were checked against rewards in the data.

2.2 Analysis

The analysis was implemented in Python following the Methods of [1] and the authors' code [3].

Trials. Left and Right trials have a first movement of that kind 125 to 400 ms after the stimulus. For the regression, the authors' published trial lists were reproduced with no differences on all 39 sessions. For Fig 2d, f, h, Fig 5a, b and Ext 6b, trials were matched for contralateral contrast across correct contra, Miss and passive trials. This keeps 34 sessions, as in the paper.

Neurons. A neuron is a unit labelled good that fires at 0.1 sp/s or more from stimulus onset to 400 ms. Regions come from the Allen label at each neuron's peak channel, merged to the paper's 42 regions through the Allen hierarchy [4]. A region needs at least 13 neurons in a session and at least 2 mice.

Firing rates. Spikes were binned at 5 ms and smoothed with a causal half-Gaussian. For display, rates are smoothed with 20 ms, averaged over trials, and smoothed with 30 ms. Normalized rates are

$$r_{\text{norm}}(t) = \frac{r(t) - r_{\text{base}}}{r_{\text{base}} + 0.5}. \tag{1}$$

Kernel regression. Each neuron's rate is modelled as a sum of kernels locked to stimuli and movements,

$$\hat{f}_n(t) = \sum_c \sum_{s \in S_c} K_{c,n}(t-s) + \sum_{m \in M} \big[K_{m,n}(t-m) + D_m K_{D,n}(t-m)\big], \tag{2}$$

with six Vision kernels, one Action kernel and one Choice kernel, giving 650 predictors. D_m is +1 for Left (ipsi) and -1 for Right (contra), so a negative pre-movement Choice kernel means contra preference. A reduced-rank basis pooled over all sessions was fit with elastic net, copying the scaling of the software used in [1] and [5]: at $\alpha = 0.5$ the penalty is $\lambda_s = 0.5/y_s$ for a neuron with rate s.d. y_s. A nested test measures what each kernel adds in cross-validation, with 2% as the threshold.

Other analyses. Population decoding used L1 logistic regression on residuals. jPECC used 10 PCs and ridge CCA at each pair of times. Choice and detect probability combined Mann-Whitney statistics across the 16 stimulus conditions, with 2,000 shuffles. Engagement used a nested ANOVA on log rates from -250 to -50 ms, and an index from the task-passive rate difference from -200 to 0 ms.

2.3 Figures

The figures rebuild from saved results, with no new analysis. Colors were measured from the high-resolution figures embedded in the paper's PDF: region colors, line colors and all colormaps. Panels were placed at the pixel positions of the original.

Three plotting problems found during the comparison were corrected, and each changed a panel's appearance substantially:

- **Fig 2d-h and Ext 4b-e:** these now use the "highly activated" neurons of the Methods' section on visualizing firing rates. On all responsive neurons, the heatmaps were far too pale.
- **Fig 4d and Ext 8:** the jPECC colormap had its black at -0.1, not 0, which turned zero correlation red.
- **Ext 5c:** the legend reads L+R and L$-$R, so the plotted lines are twice each kernel.

Ext 1 was also found to use non-repeat trials. Using `repNum = 1` gives the paper's trial count exactly for 9 of 10 subjects.

3 Assignment figures: VISp, LGd and LP

These figures redo Fig 2a-c, Fig 4e, Fig 5a and Ext 6b for the three regions, in the paper's style.

Region	Neurons	Sessions	Responsive	Regression	Pass 2%	Vision	Action	Choice	T−P	G−M
VISp	917	12	616	708	259	208	16	1	-0.68	+0.02
LGd	789	8	362	481	33	4	11	0	-1.67	-0.36
LP	680	9	344	401	78	22	17	0	-1.29	-0.25

Table 2: The three regions in the full run. Vision, Action and Choice count neurons passing the 2% nested test. T−P is the pre-stimulus rate difference, task minus passive, and G−M is Go minus Miss, both in sp/s.

3.1 Fig 2a-c style: single-neuron rasters

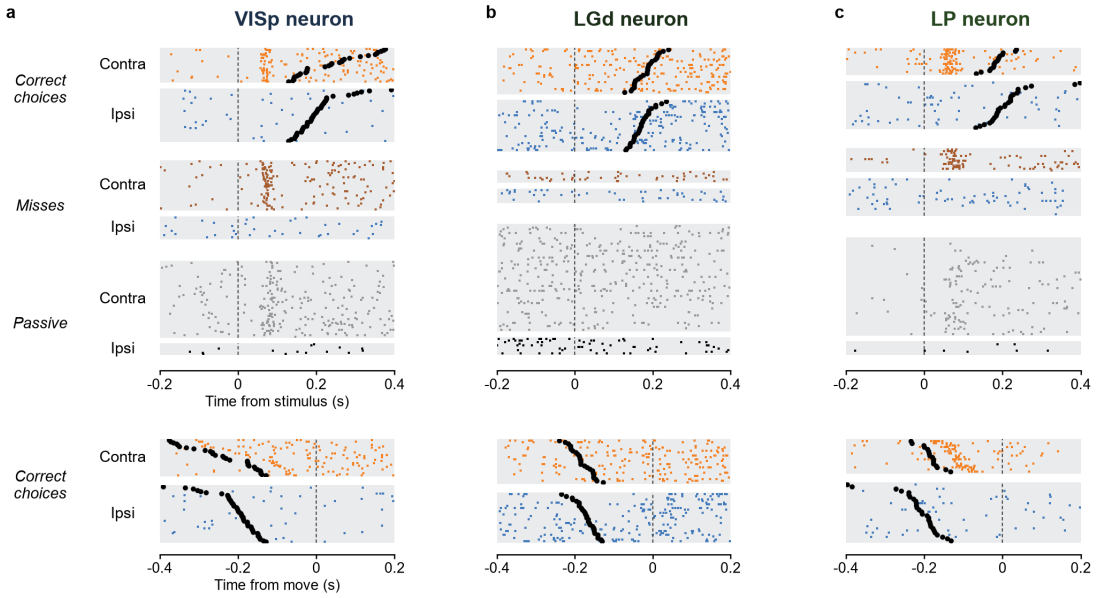


Figure 1: One neuron per region, on trials with a stimulus on one side only. Top six rows are aligned to the stimulus, with black dots marking movement. Bottom two rows are aligned to movement, with black dots marking the stimulus.

The VISp neuron fires a burst about 70 ms after a contralateral stimulus on correct, Miss and passive trials, and not after ipsilateral stimuli. The response is present with no movement at all, so it is visual. The LP neuron shows the same pattern. The LGd neuron is different: it fires at a high rate throughout, with little locked to the stimulus. This fits the paper's report that recorded LGd neurons had receptive fields away from the task stimuli [1].

3.2 Fig 4e style: firing aligned to movement

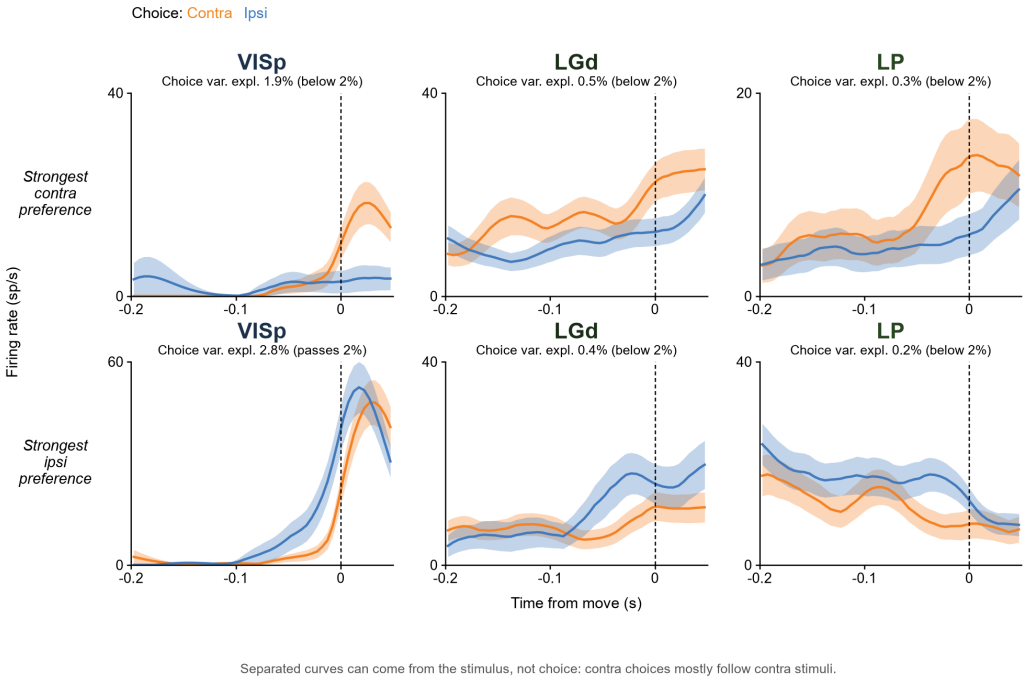


Figure 2: Rates aligned to movement, for contralateral (orange) and ipsilateral (blue) choices. Neurons have the highest nested Choice variance explained among those firing at 8 sp/s or more, in sessions with at least 15 trials per side.

Only one of the six neurons passes the 2% Choice threshold: the VISp neuron in the bottom row, which is the single neuron of 708 in VISp to do so. No LGd or LP neuron passes. This agrees with the paper, which found no choice coding in VISp [1], and one neuron of the 259 tested (0.4%) is no more than the 0.5% false-positive rate of the shuffle test. Separated curves are therefore not evidence of choice coding here. Contra choices mostly follow contra stimuli, so a visual response alone can separate the two curves. The same caution applies to the VISp neuron in the paper's Fig 4a.

3.3 Fig 5a style: Miss against Passive

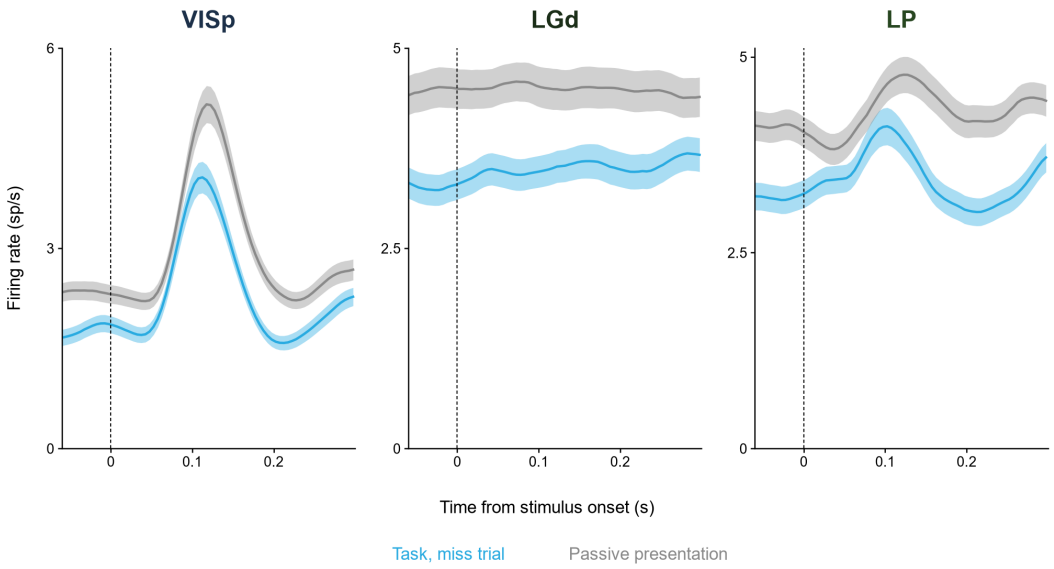


Figure 3: Population average rate on Miss and passive trials with matched contralateral contrast. Shading is s.e.m. across neurons.

All three regions fire less before the stimulus in the task than in passive replay, by 0.68 sp/s in VISp, 1.67 in LGd and 1.29 in LP. The nested ANOVA finds this significant in all three. The paper reports the same direction for neocortex and visual thalamus [1]. VISp also shows the visual response after the stimulus, with a similar shape in the two conditions.

3.4 Ext 6b style: Go, Miss and Passive

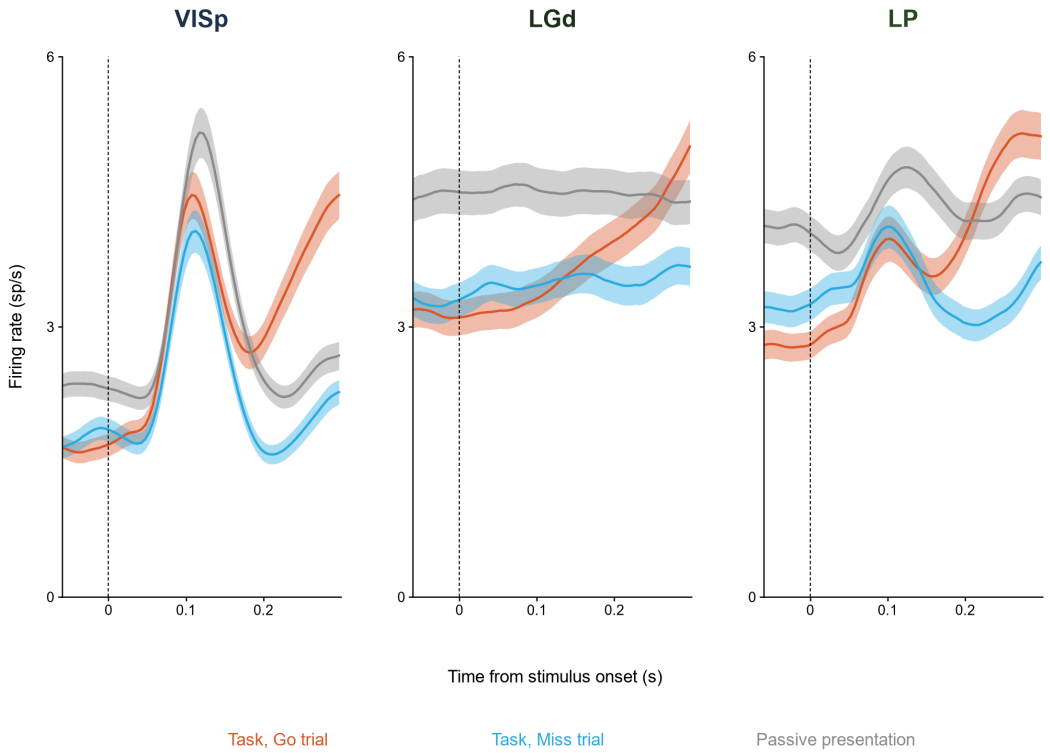


Figure 4: Population average rate on Go, Miss and passive trials with matched contralateral contrast.

Before the stimulus, the order is Passive above Miss and Go in all three regions, which is the pattern the paper describes for these regions [1]. Go and Miss are close before the stimulus in VISp, and Go is a little below Miss in LGd and LP. After about 150 ms the Go trace rises, reflecting movement.

3.5 Neurons shown

Region	Panel	Session	Cluster	Choice var. expl.
VISp	Raster	`Muller_2017-01-07`	683	-
VISp	Contra preference	`Forssmann_2017-11-01`	268	1.9%
VISp	Ipsi preference	`Cori_2016-12-18`	798	2.8%
LGd	Raster	`Lederberg_2017-12-09`	454	-
LGd	Contra preference	`Lederberg_2017-12-09`	454	0.5%
LGd	Ipsi preference	`Forssmann_2017-11-01`	534	0.4%
LP	Raster	`Hench_2017-06-15`	1244	-
LP	Contra preference	`Moniz_2017-05-15`	179	0.3%
LP	Ipsi preference	`Moniz_2017-05-15`	235	0.2%

Table 3: Neurons in the assignment figures. Raster neurons have the most reliable contralateral stimulus response in a session with at least 15 correct trials per side.

4 Example neurons

The paper does not identify its example neurons. To get as close as the data allow, the curves in each example panel were digitized from the paper's high-resolution figures, and every neuron in that region was scored against them in sp/s. Candidates had to pass the test their panel implies: a neuron shown as choice-selective must pass the nested Choice test, and a neuron shown with no visual response must not pass the Vision test. For Fig 2, row heights also count, because they follow each session's trial counts.

Curve error is the root-mean-square difference from the paper's curves, as a percentage of the panel's peak. The chosen neurons are close matches, and they need not be the same cells. One result stands out: for the VISpm neuron of Fig 2b, curves and row heights separately pointed to `Cori_2016-12-17`, so that may be the paper's own session.

Panel	Region	Session	Cluster	Curve error
Fig 2a, 3b	SUB	`Hench_2017-06-15`	1205	30%
Fig 2b, 3a	VISpm	`Cori_2016-12-17`	1145	16%
Fig 2c, 4a (ZI)	ZI	`Richards_2017-11-01`	791	9%
Fig 4a VISp	VISp	`Cori_2016-12-18`	647	15%
Fig 4a MRN	MRN	`Radnitz_2017-01-11`	420	10%
Fig 4a CP	CP	`Radnitz_2017-01-12`	1105	6%
Ext 4a VISp	VISp	`Moniz_2017-05-16`	1096	4%
Ext 4a VISam	VISam	`Muller_2017-01-08`	605	13%
Ext 5a, c	APN	`Moniz_2017-05-15`	669	19%
Fig 4e panel 1	MRN	`Muller_2017-01-07`	969	24%
Fig 4e panel 2	MRN	`Radnitz_2017-01-11`	426	14%
Fig 4e panel 3	SCm	`Tatum_2017-12-07`	665	5%
Fig 4e panel 4	SCm	`Lederberg_2017-12-07`	820	4%
Fig 4e panel 5	MOs	`Muller_2017-01-07`	423	21%
Fig 4e panel 6	MOs	`Radnitz_2017-01-08`	467	6%
Fig 4e panel 7	CP	`Richards_2017-11-01`	302	26%
Fig 4e panel 8	CP	`Hench_2017-06-18`	440	17%

The top five candidates for each panel are in `analysis/results/example_scores.csv`.

5 Comparison with the paper's numbers

Quantity	Paper	Full run
Neurons	29134	29178
Localized neurons	22458	22210
Modulated neurons	13,466 (60.0%)	12,972 (58.4%)
Sessions with contrast-matched trials	34 of 39	34 of 39
Sessions in CP analysis	33 of 39	33 of 39
Sessions in DP analysis	not stated	39 of 39
Shuffle false positives	0.33%	0.50%
Midbrain choice neurons preferring contra	53/54	45/46
Midbrain suppressed for non-preferred choice	29/54	16/46
Forebrain choice neurons preferring ipsi	19/48	11/33
Forebrain suppressed for non-preferred choice	10/48	6/33
Focality: action, contra vision, choice	0.040, 0.069, 0.079	0.040, 0.065, 0.071
Fig 5d correlation	r = 0.46, p = 0.002	r = 0.58, p = 6.33e-05

Quantity	Paper	Full run
Mean engagement index difference	8.42	9.15 (n = 34)
Ext 7f with Go-Miss, DP-qualified sessions (39)	r = 0.69	r = 0.79
Ext 7f with Task-Passive, DP-qualified sessions	r = 0.48	r = 0.50
Ext 7f with Go-Miss, 33-session CP subset sessions (33)	r = 0.69	r = 0.81
Ext 7f with Task-Passive, 33-session CP subset sessions	r = 0.48	r = 0.47

Exact equality is not expected for results that depend on random splits, or on a different solver from the original.

Region	Paper	Ours	%	Region	Paper	Ours	%	Region	Paper	Ours	%
ACA	608	651	+7.1	MD	381	388	+1.8	SCm	997	992	-0.5
ACB	255	261	+2.4	MG	276	284	+2.9	SCs	317	281	-11.4
APN	235	236	+0.4	MOp	682	680	-0.3	SNr	274	273	-0.4
BLA	261	266	+1.9	MOs	1534	1335	-13.0	SSp	461	448	-2.8
CA1	1129	1150	+1.9	MRN	857	875	+2.1	SUB	669	661	-1.2
CA3	438	430	-1.8	OLF	684	695	+1.6	VISa	285	288	+1.1
CP	914	910	-0.4	ORB	770	789	+2.5	VISam	805	804	-0.1
DG	711	702	-1.3	PAG	130	128	-1.5	VISl	403	382	-5.2
GPe	274	281	+2.6	PL	728	730	+0.3	VISp	923	917	-0.7
ILA	338	349	+3.3	PO	620	645	+4.0	VISpm	516	494	-4.3
LD	308	275	-10.7	POL	190	186	-2.1	VISrl	236	220	-6.8
LGd	811	789	-2.7	POST	272	272	+0.0	VPL	255	294	+15.3
LP	732	680	-7.1	RSP	598	581	-2.8	VPM	357	327	-8.4
LS	844	846	+0.2	RT	160	152	-5.0	ZI	220	263	+19.5

Table 6: Neurons per region against Extended Data Table 1. The total is within 2%, but the released labels do not reproduce the table region by region. Re-deriving labels from the atlas at each channel's position was much worse (total error of 8,505 neurons against 790), so the released labels were kept.

6 Known differences from the paper

- Elastic net solved at lambda directly; the authors interpolate along a glmnet path.
- Example neurons are the real neurons closest to the paper's panels; the paper does not identify its examples, so they need not be the same cells.
- Fig 2d-h heatmaps and Ext 4b-e use the Methods' highly activated neurons (sliding rank-sum); the captions say responsive.
- Ext 1 uses non-repeat trials (repNum 1), which reproduces the paper's trial counts for 9 of 10 subjects. Cori has 536 against 526.
- Panels with no data behind them (photos, histology, cartoons) are crops of the paper's own figure, tagged in the figure.
- jPECC averages use every session with both region groups: 17, 11 and 9 sessions against the paper's 10, 5 and 9. No neuron-count rule reproduced those numbers.
- Fig 1j uses a session with six of the seven regions of the paper's panel.
- Released region labels differ from the paper's table per region (table below).
- Pupil panels use all 39 sessions; the paper's five rejected recordings are not named.
- Fig 2d-h and Fig 5b use two-sided $|z|>4$ as in the Methods; captions read as one-sided. One-sided counts are below.

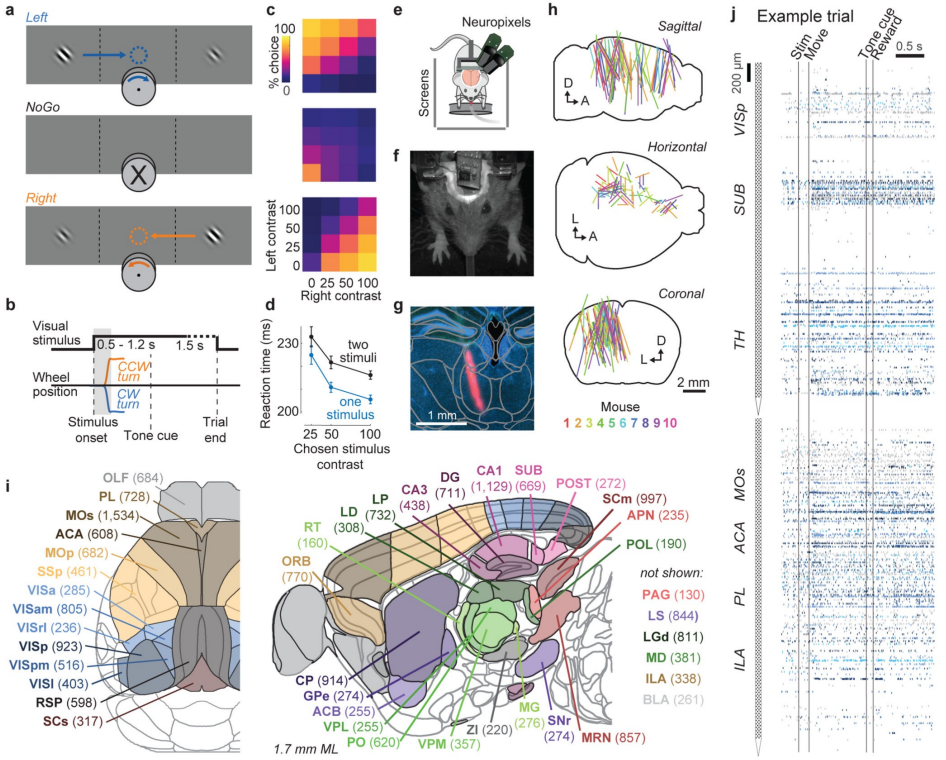
- Contrast-matching Go group is correct contra trials, which gives the paper's 34 sessions; the Methods text says Left or Right.
- Fig 4c normalizes by group, reading the Methods' 'area' as the three groups.
- Decoding regularization and folds are not stated: L1 logistic regression, C from (0.01, 0.1, 1), 5 outer and 3 inner folds.
- ccCP counts use a 100 ms box stepped by 10 ms; the time-course window is not stated.
- Ext 4b latency is when the odd-trial rate first reaches half its peak; its definition is not stated.
- Ext 9e-f and 9k use within-session median splits; the paper gives labels only.
- Fig 5c-d and Ext 7f use -0.2 to 0 s; their windows are not stated.
- The 34 engagement sessions are taken as the contrast-matched sessions; the paper does not say which 34.
- Fig 1d ANOVA uses chosen contrast, stimulus count and subject; its factors are not stated.
- Passive trials in matching require no movement from -0.05 to 0.4 s; this does not change the session count.
- DP applies the >=10 includable-trials rule itself, so it keeps more sessions than CP.
- Ext 5f uses neurons passing the 2% full-model cut, matching the neuron counts implied by its histograms.
- Ext 5g's color scale is set from the data, because the paper's projection units are not stated.
- Ext 5c plots the sum and difference of the Left and Right responses, as the paper's legend labels them: twice each kernel.
- Released region labels are kept: labels read from the atlas at channel positions were far further from Extended Data Table 1.
- Region, line and colormap colors are measured from the paper's figures. BLA-style pale regions and the brain-map color steps are approximate.

7 Figure-by-figure comparison

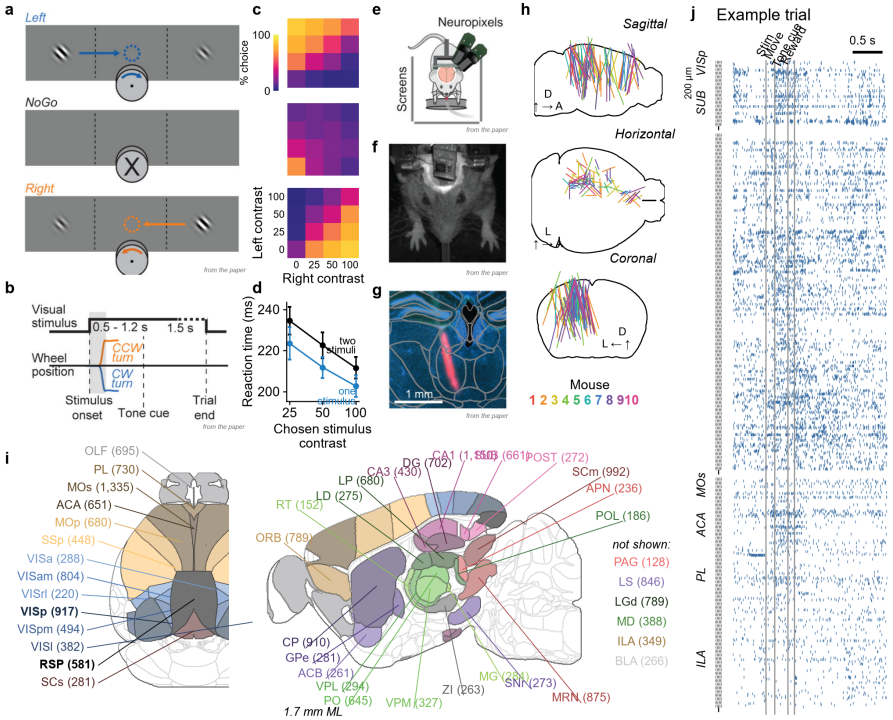
Each page shows the paper's figure beside ours, followed by a table for every panel. Ratings are defined in Section 1.

Figure 1

Original



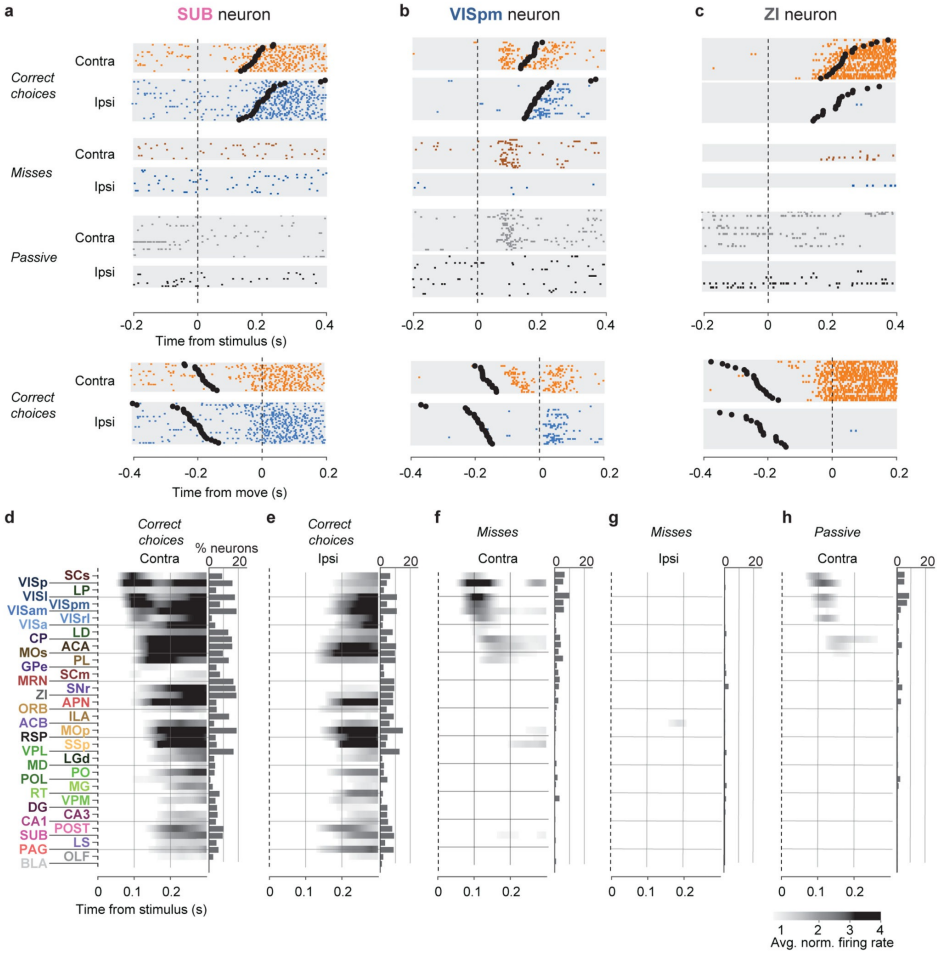
Reproduced from the full 39-session run



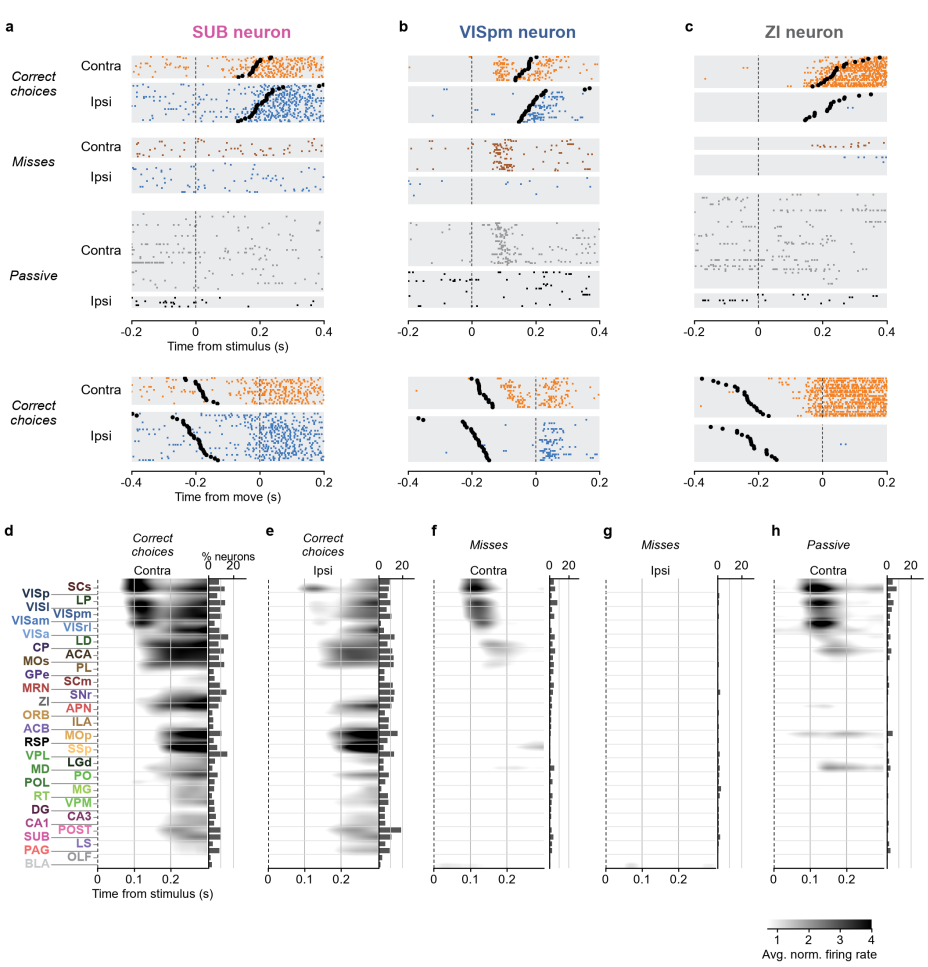
Panel	Agreement	What the comparison shows
a	From the paper	Task cartoon.
b	From the paper	Trial timeline cartoon.
c	Close	Same choice pattern for every contrast pair. Colormap measured from the paper.
d	Close	Reaction times of 200 to 240 ms, slower with two stimuli and with low contrast. ANOVA p = 2.6e-10 (paper p < 1e-4).
e	From the paper	Setup cartoon.
f	From the paper	Photograph.
g	From the paper	Histology.
h	Close	Probe tracks from each channel's CCF position, with the same spread. Mouse colors are approximate.
i	Close	Same map and labels. Counts are ours: 22,210 localized neurons against 22,458, with 23 of 42 regions more than 2% off.
j	Partial	A real trial, from a session holding six of the panel's seven regions. It has no TH label, so the layout differs.

Figure 2

Original

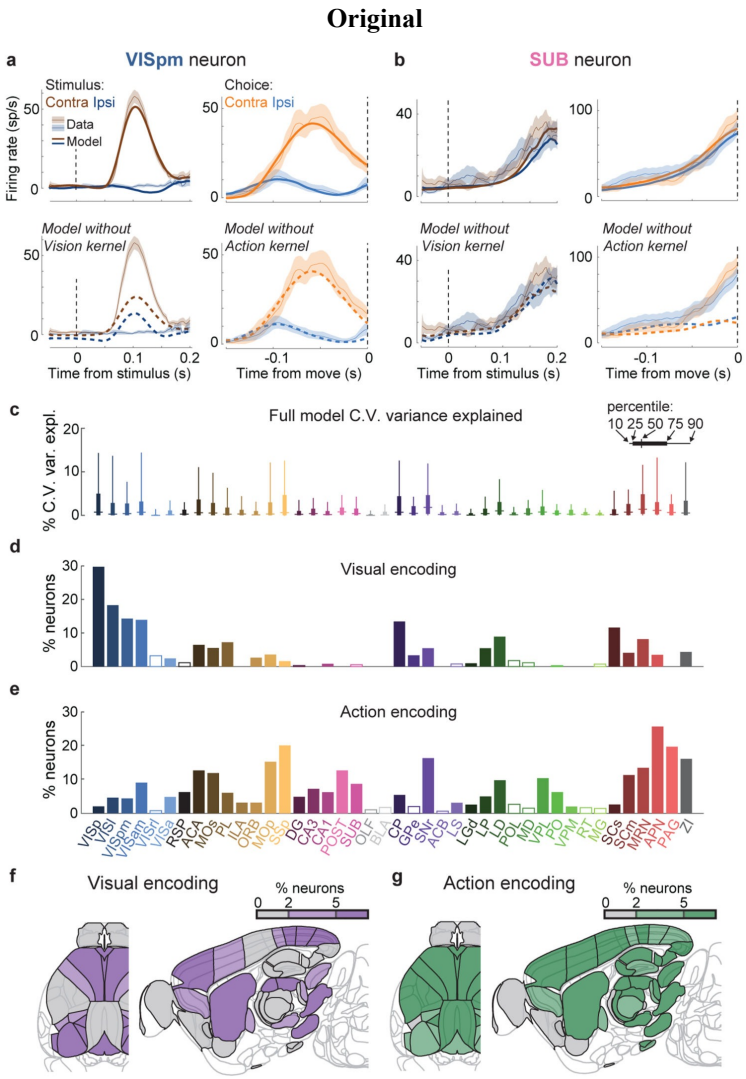


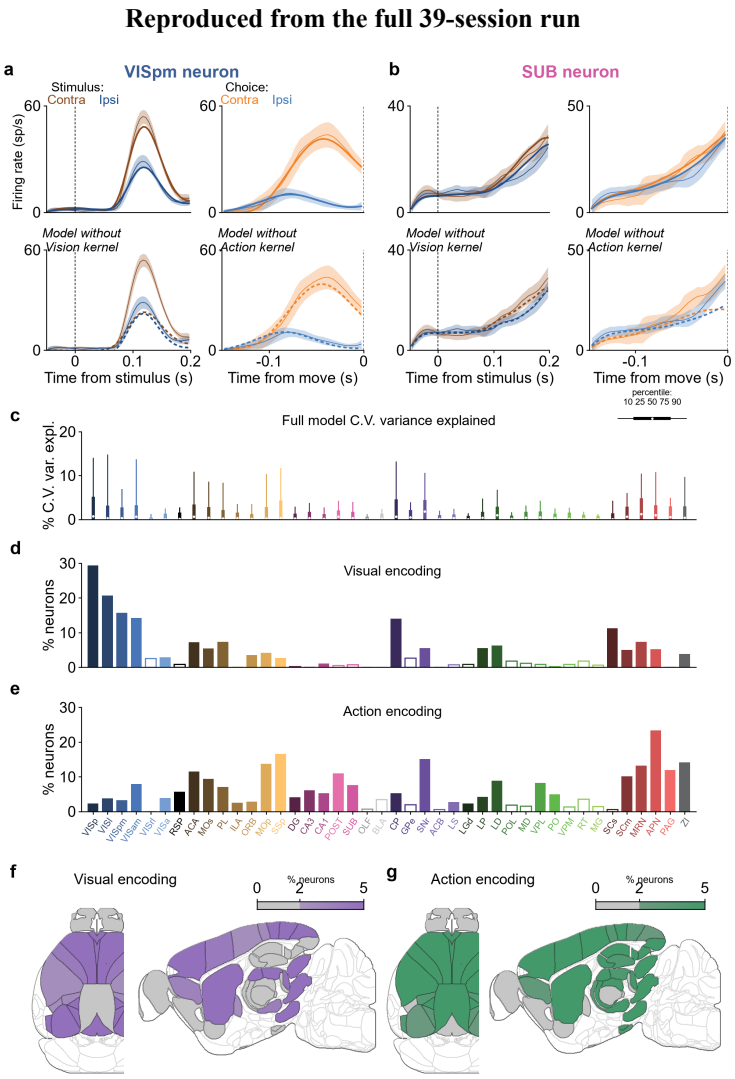
Reproduced from the full 39-session run



Panel	Agreement	What the comparison shows
a	Close	SUB neuron firing before movement to both sides, with no stimulus response. Row heights differ, so this is probably not the same cell.
b	Close	VISpm neuron with a contra stimulus burst near 0.1 s and later movement firing. Curves and row heights independently point to Cori_2016-12-17.
c	Close	ZI neuron firing before contra choices only.
d	Close	Visual areas first, then spread to most regions.
e	Close	Most regions active, with visual areas late.
f	Close	Activity stays in the visual pathway.
g	Close	Almost no activity.
h	Partial	Visual pathway only, as in the paper, but stronger than Miss in visual cortex where the paper's is weaker.

Figure 3

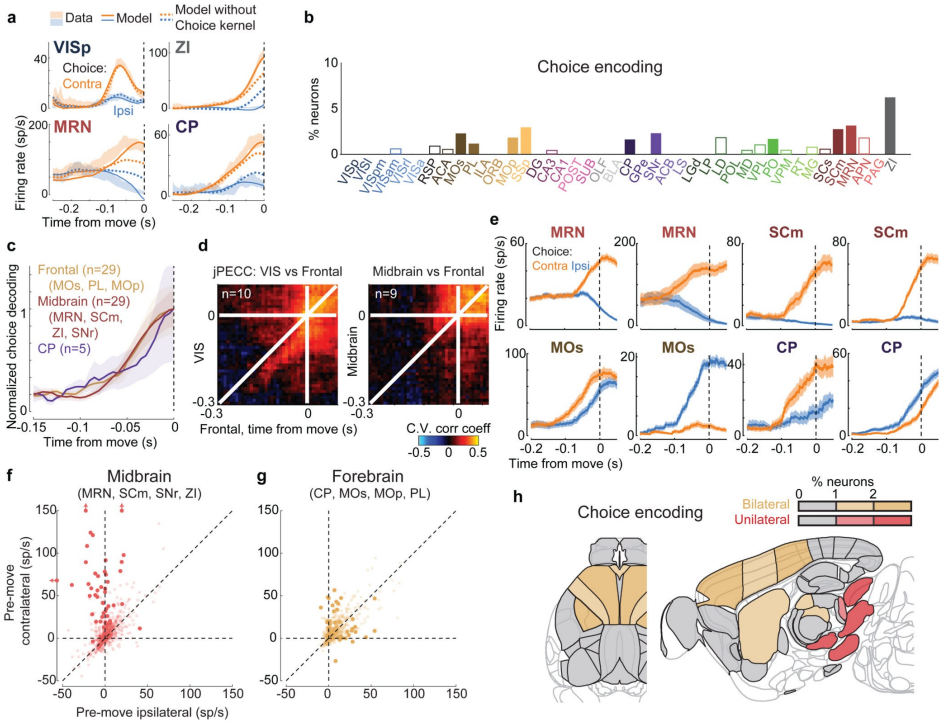

Original
a
VISpm neuron
b
SUB neuron
Stimulus:
Contra Ipsi
Data
Model
Choice:
Contra Ipsi
Firing rate (sp/s)
Model without Vision kernel
Model without Action kernel
Time from stimulus (s)
Time from move (s)
c
Full model C.V. variance explained
percentile:
10 25 50 75 90
% C.V. var. expl.
d
Visual encoding
% neurons
e
Action encoding
f
Visual encoding
% neurons
g
Action encoding


Reproduced from the full 39-session run
a
VISpm neuron
b
SUB neuron
Stimulus:
Contra Ipsi
Choice:
Contra Ipsi
Firing rate (sp/s)
Model without Vision kernel
Model without Action kernel
Time from stimulus (s)
Time from move (s)
c
Full model C.V. variance explained
percentile:
10 25 50 75 90
% C.V. var. expl.
d
Visual encoding
% neurons
e
Action encoding
f
Visual encoding
% neurons
g
Action encoding

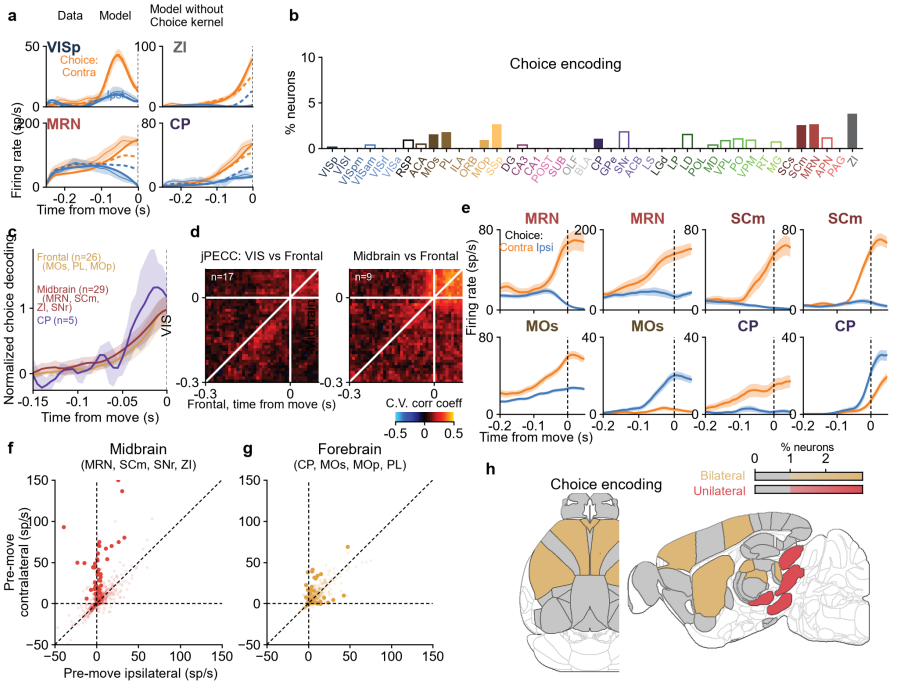
Panel	Agreement	What the comparison shows
a	Close	Model fits and the loss of fit without the Vision kernel both match. Vision peak about 50 sp/s.
b	Close	Good fit is lost without the Action kernel, as in the paper. Peak rate is about 40 sp/s against 80.
c	Same	Same distribution across regions.
d	Close	Same regions. VISp 29.4% against about 30%.
e	Close	Broad action coding across regions. APN 23.4% against about 25%.
f	Close	Same map; the color steps are approximated.
g	Close	Same map; the color steps are approximated.

Figure 4

Original

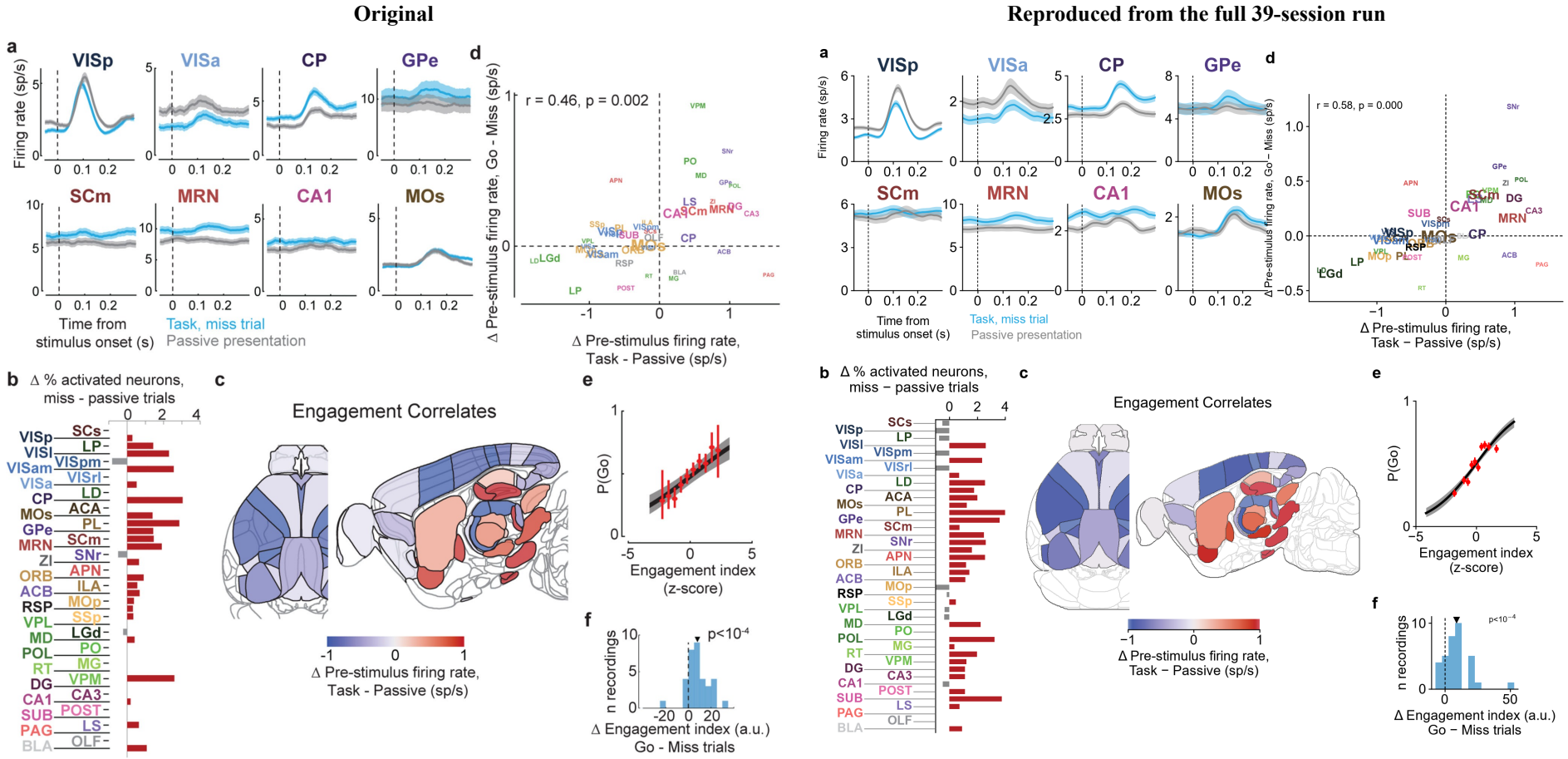


Reproduced from the full 39-session run



Panel	Agreement	What the comparison shows
a	Close	VISp fits without Choice; ZI, MRN and CP need it.
b	Close	Same regions. ZI 3.8% against about 6%, SCm 2.6% against about 3%.
c	Partial	Similar time course and no time-by-group interaction ($p = 1.0$). However, the group main effect is significant here ($p = 7e\text{-}4$) and not in the paper. Frontal $n = 26$ against 29.
d	Close	VIS leads frontal, and midbrain and frontal show no lag. VIS panel uses $n = 17$ against 10.
e	Close	All eight panels show the same shape and scale as the paper. Error 4% to 26% of the panel's peak.
f	Close	45 of 46 midbrain choice neurons prefer contra, against 53 of 54.
g	Close	11 of 33 forebrain choice neurons prefer ipsi, against 19 of 48.
h	Close	Same bilateral and unilateral regions.

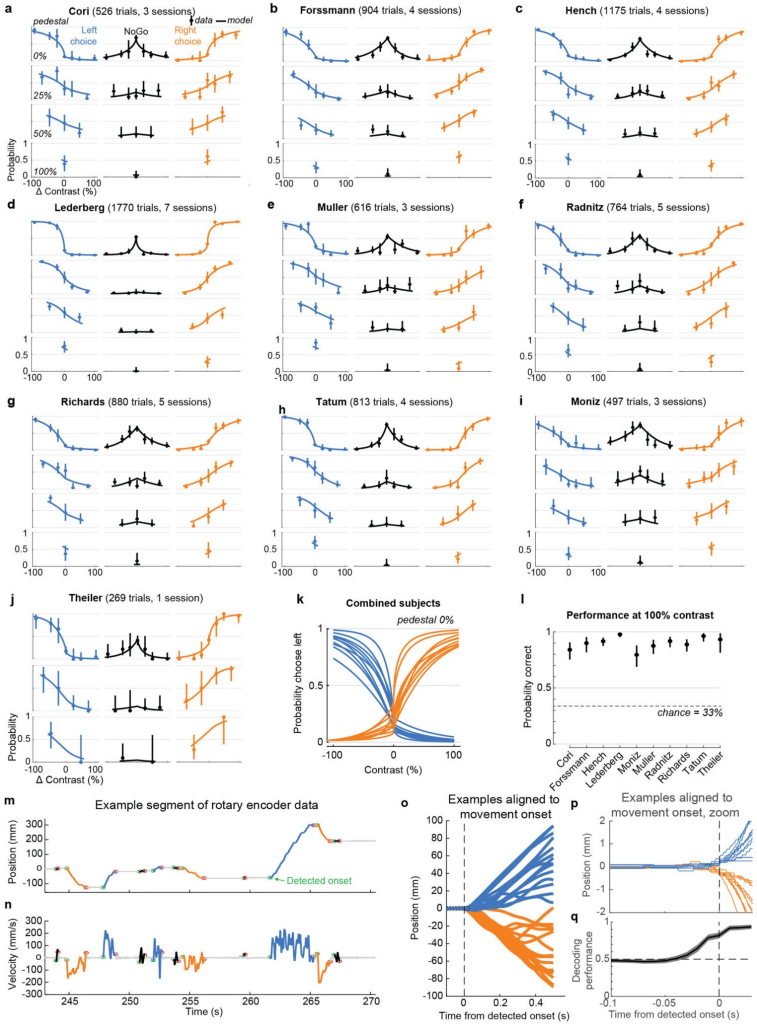
Figure 5


Original
Reproduced from the full 39-session run
a
VISp
VISa
CP
GPe
SCm
MRN
CA1
MOs
Firing rate (sp/s)
Time from stimulus onset (s)
Task, miss trial
Passive presentation
d
r = 0.46, p = 0.002
r = 0.58, p = 0.000
Δ Pre-stimulus firing rate, Go - Miss (sp/s)
Δ Pre-stimulus firing rate, Task - Passive (sp/s)
b
Δ % activated neurons, miss - passive trials
c
Engagement Correlates
Δ Pre-stimulus firing rate, Task - Passive (sp/s)
e
P(Go)
Engagement index (z-score)
f
n recordings
p<10-4
Δ Engagement index (a.u.)
Go - Miss trials

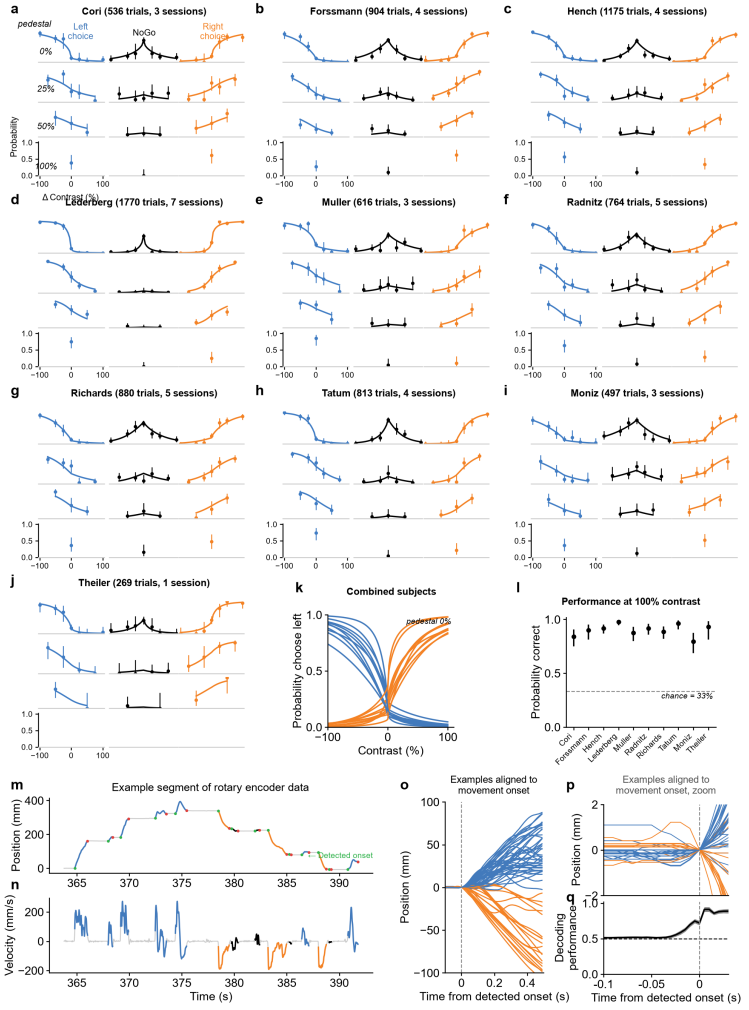
Panel	Agreement	What the comparison shows
a	Close	Pre-stimulus Miss minus Passive has the paper's sign in all eight regions, though GPe and SCm are within their error. Contrast matched.
b	Partial	Mostly positive as in the paper, but several regions differ in size.
c	Close	Neocortex and visual thalamus below passive, subcortex above.
d	Close	Same positive relation, r = 0.58 against 0.46. SNr is highest here, VPM in the paper.
e	Close	P(Go) rises with the engagement index.
f	Close	Mean 9.15 against 8.42, with one outlying recording near 50.

Extended Data Figure 1

Original



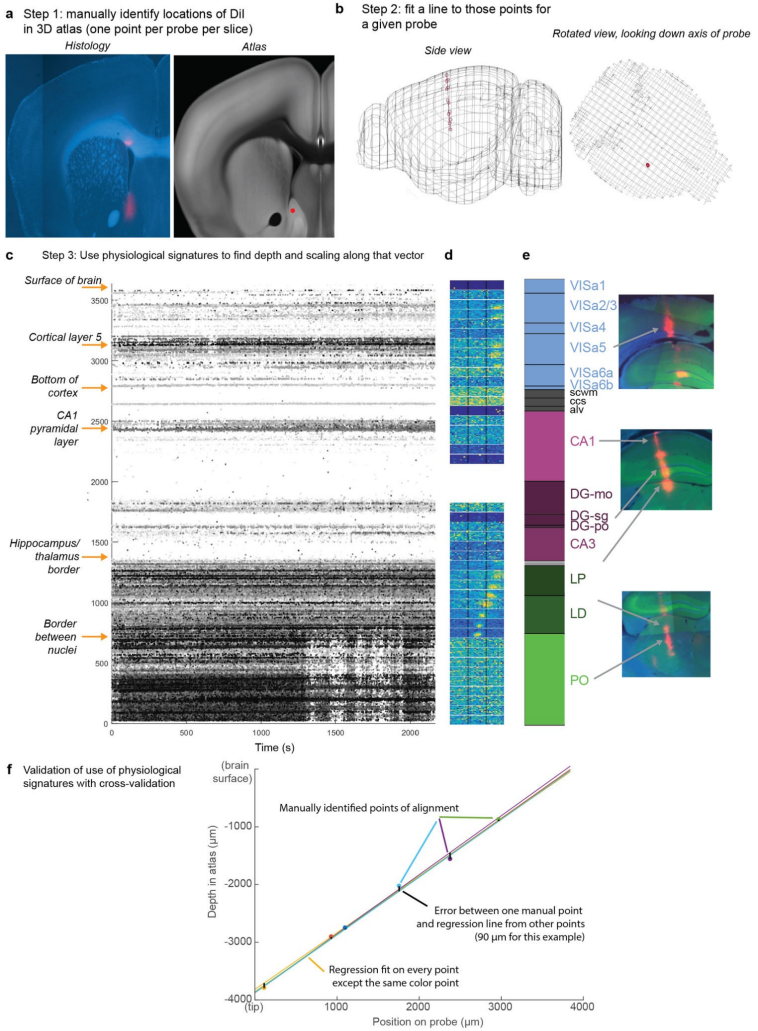
Reproduced from the full 39-session run



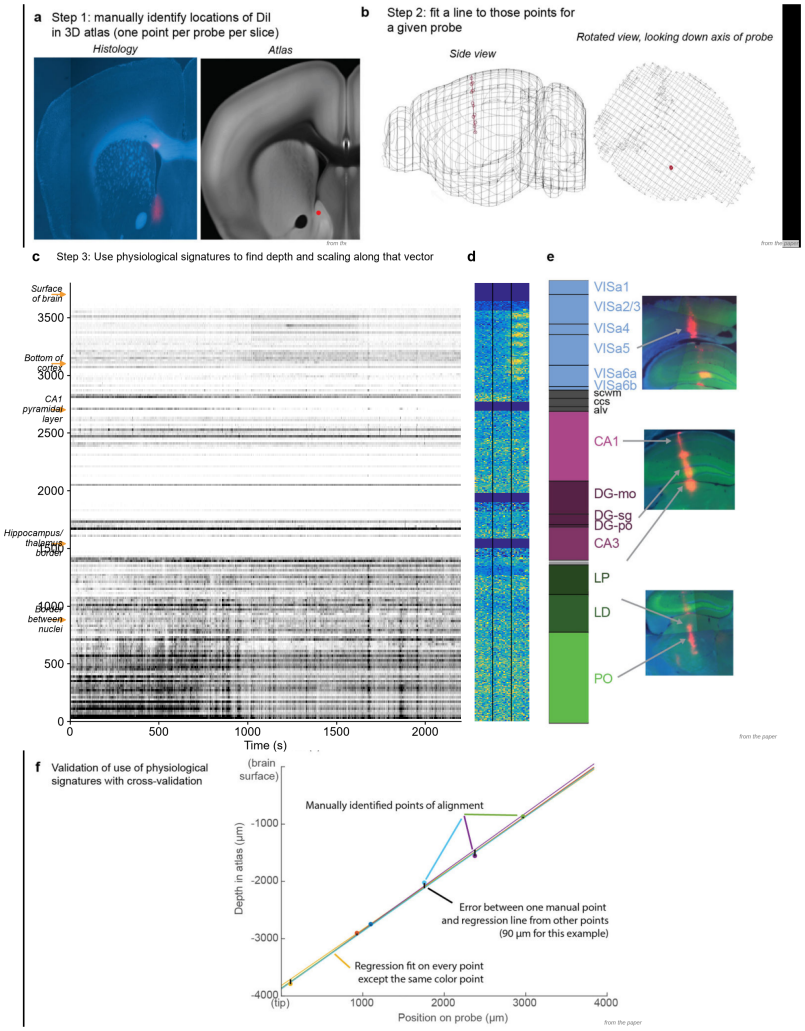
Panel	Agreement	What the comparison shows
a-j	Same	Non-repeat trials give the paper's trial counts for 9 of 10 subjects (Cori 536 against 526). Fits overlie the data as in the paper.
k	Close	Same family of fits.
l	Close	Same performance by subject.
m, n	Close	A real wheel segment, chosen automatically, so its shape differs.
o	Close	Same aligned turns.
p	Partial	Same onset, with more small pre-onset excursions than the paper.
q	Close	Direction decoding at chance before onset, rising through it.

Extended Data Figure 2

Original



Reproduced from the full 39-session run

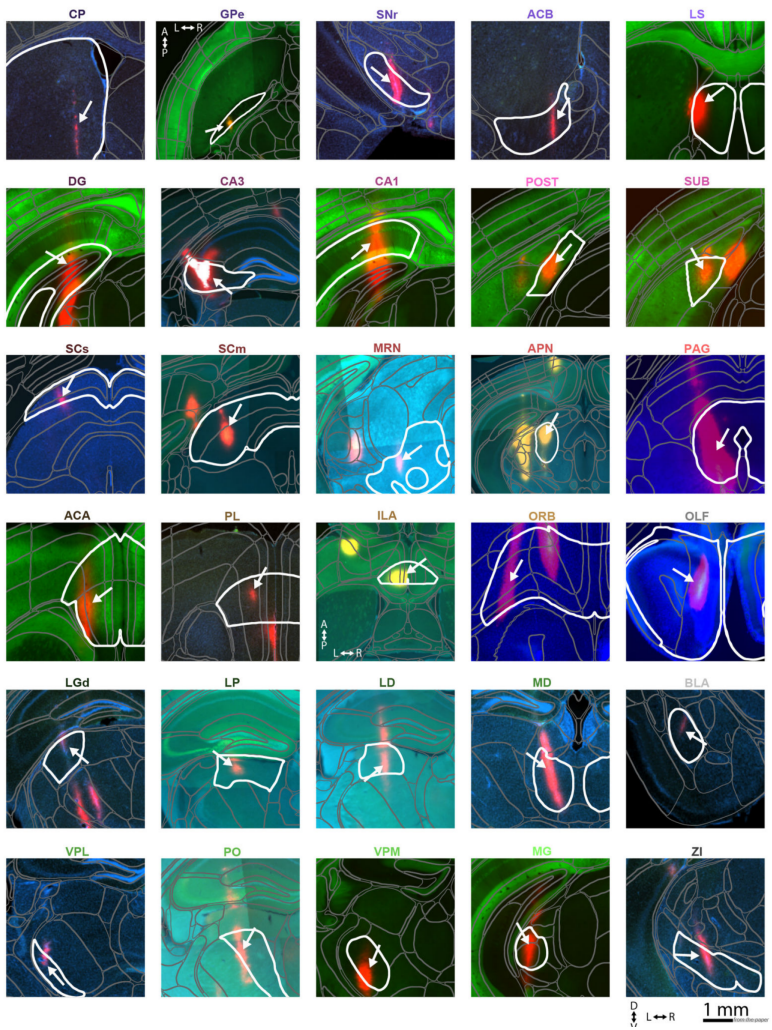


Panel	Agreement	What the comparison shows
a, b	From the paper	Histology and 3D track fit.
c	Close	Real spike raster from the one probe passing through all the paper's regions. Landmark labels come from that probe's region borders.
d	Close	Real receptive-field maps from sparse noise on the same probe.
e	From the paper	Histology.
f	From the paper	Alignment landmarks are not in the dataset.

Extended Data Figure 3

Original

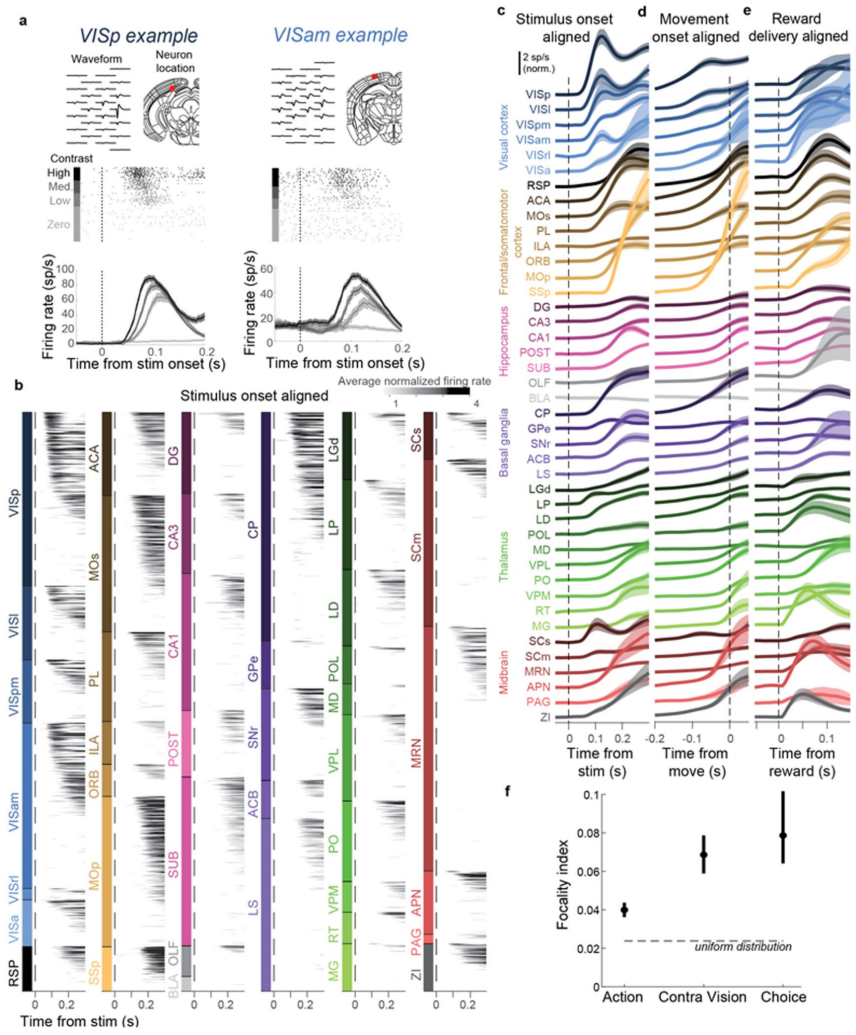
Reproduced from the full 39-session run



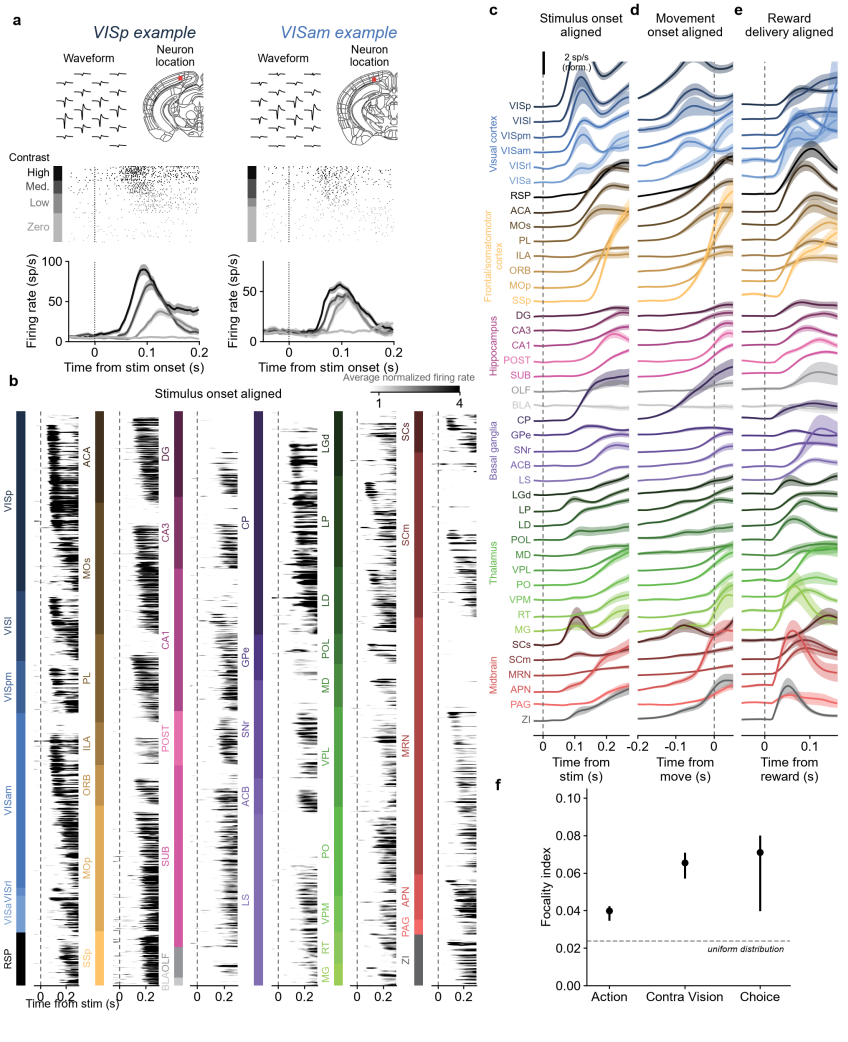
Panel	Agreement	What the comparison shows
all	From the paper	Histology.

Extended Data Figure 4

Original



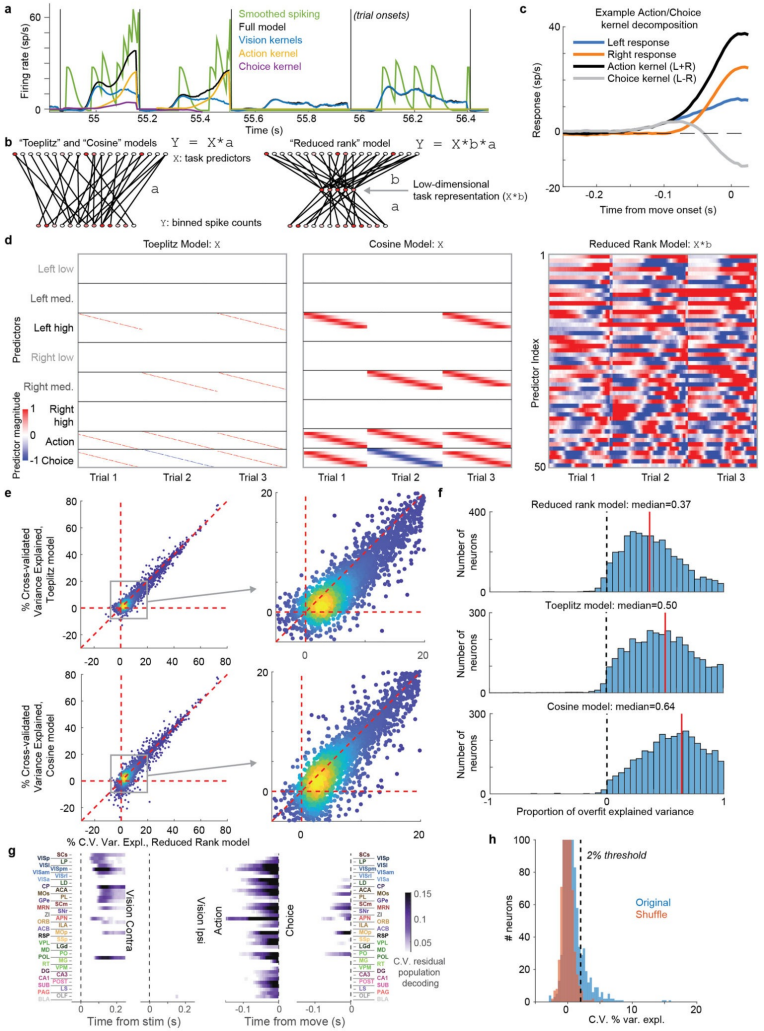
Reproduced from the full 39-session run



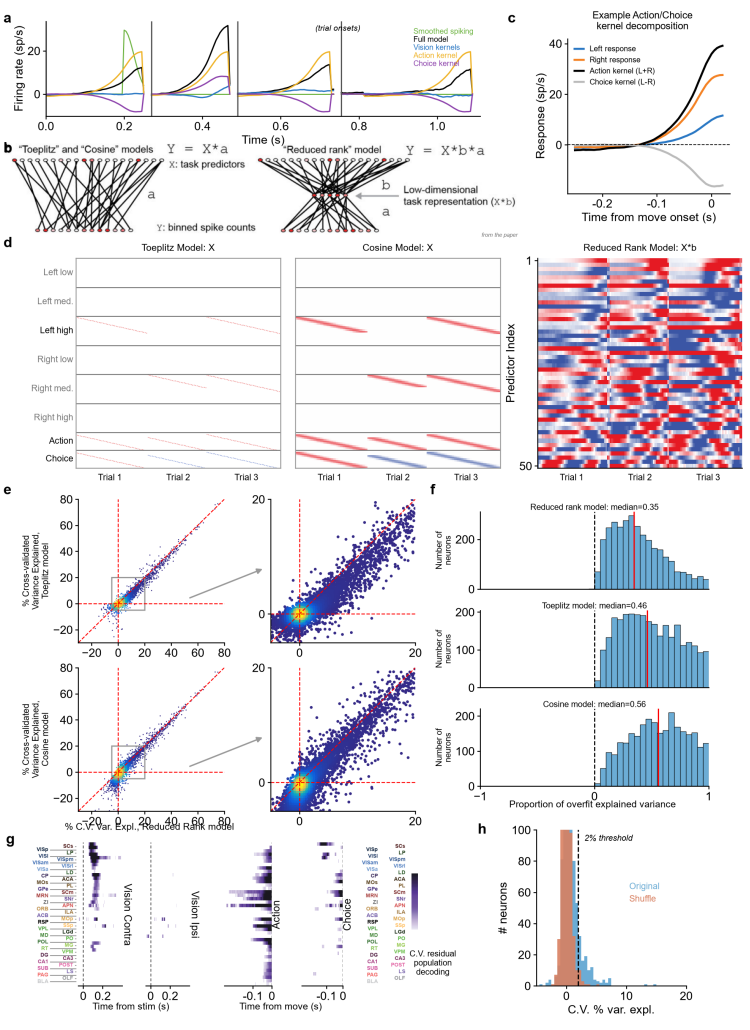
Panel	Agreement	What the comparison shows
a	Close	Contrast tuning, peak rates and rasters close to the paper. Error 4% and 13%.
b	Partial	Latency sorted, odd trials sorting even trials, with region strips in place. It is visibly denser than the paper's.
c-e	Close	Same shape per region. Visual areas first, wide spread by movement, and reward responses.
f	Close	0.040, 0.065 and 0.071 against 0.040, 0.069 and 0.079.

Extended Data Figure 5

Original



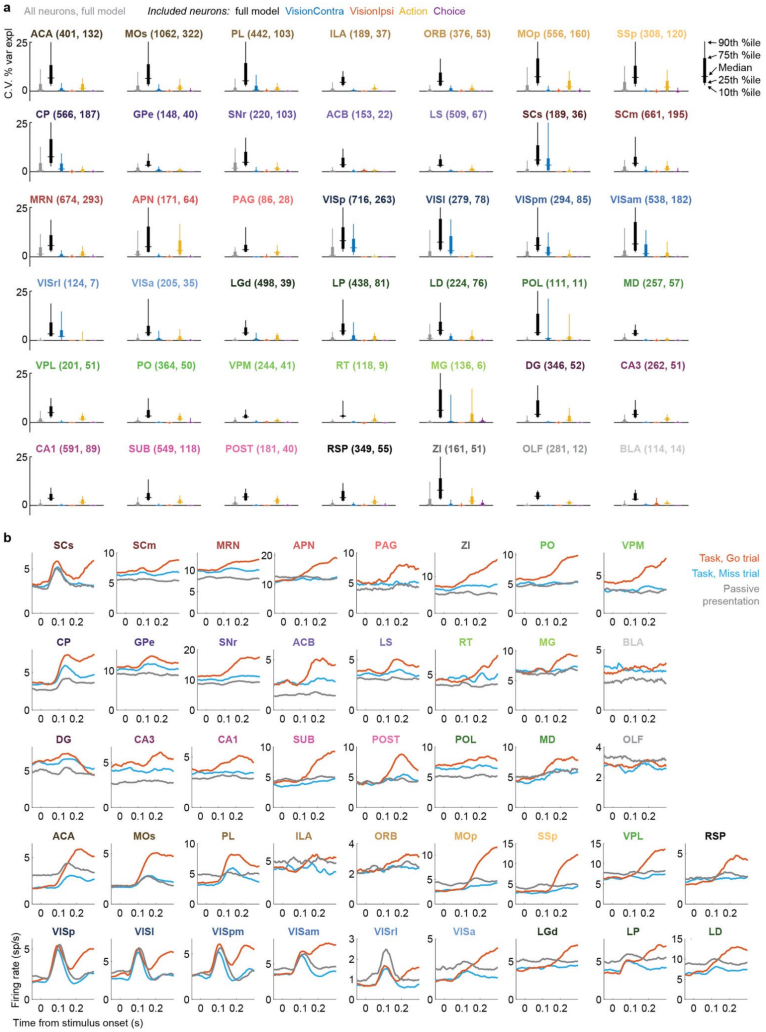
Reproduced from the full 39-session run



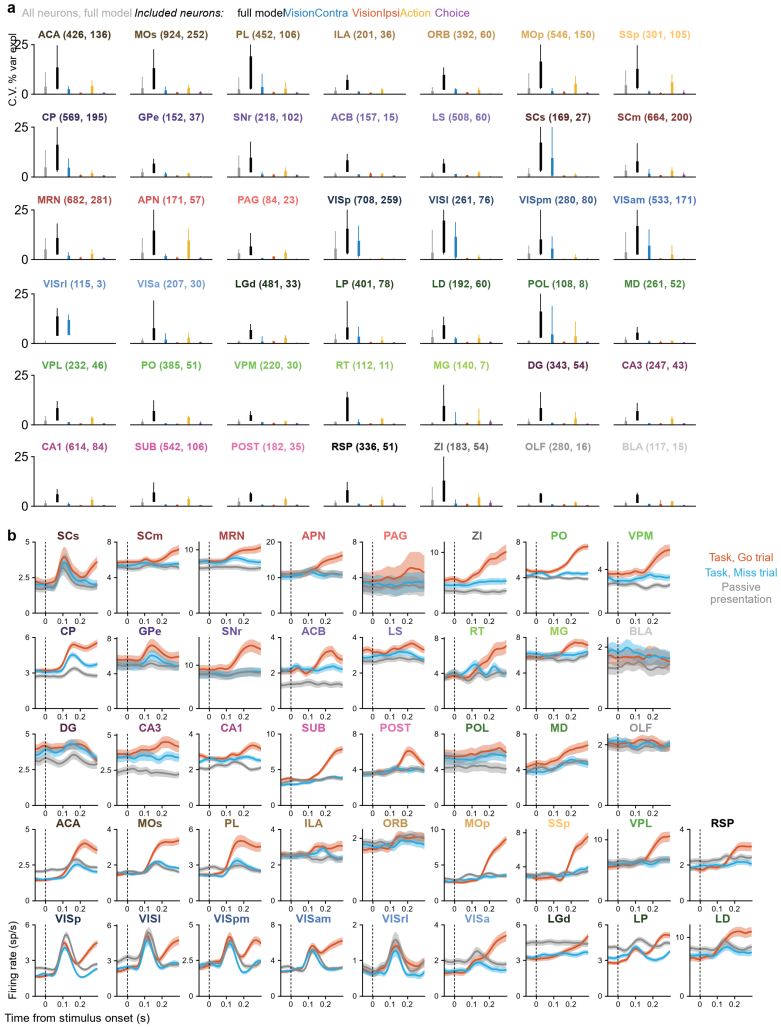
Panel	Agreement	What the comparison shows
a	Close	Kernel decomposition of a real neuron on four trials.
b	From the paper	Model cartoons.
c	Close	Left 12 and Right about 28 sp/s against 12 and 25.
d	Close	Same predictor structure.
e	Close	Reduced rank beats both other models.
f	Close	Medians 0.35, 0.46 and 0.56 against 0.37, 0.50 and 0.64.
g	Partial	Same regions decode. The color scale is set from the data, because the paper's units are not stated.
h	Close	0.50% false positives against 0.33%.

Extended Data Figure 6

Original



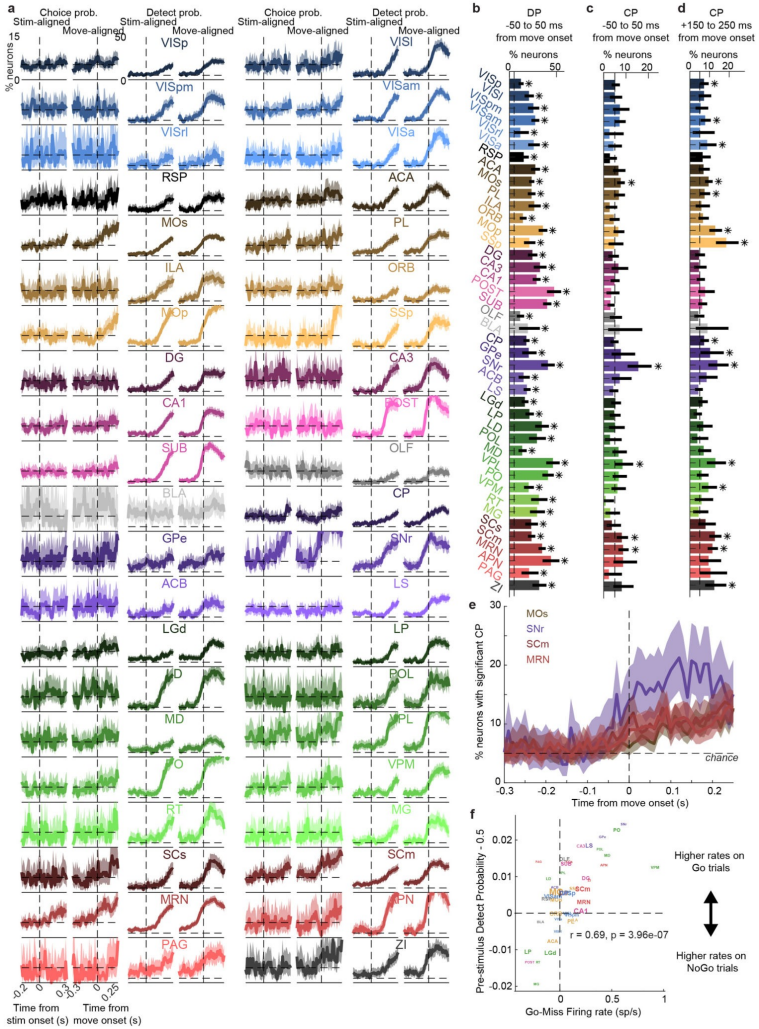
Reproduced from the full 39-session run



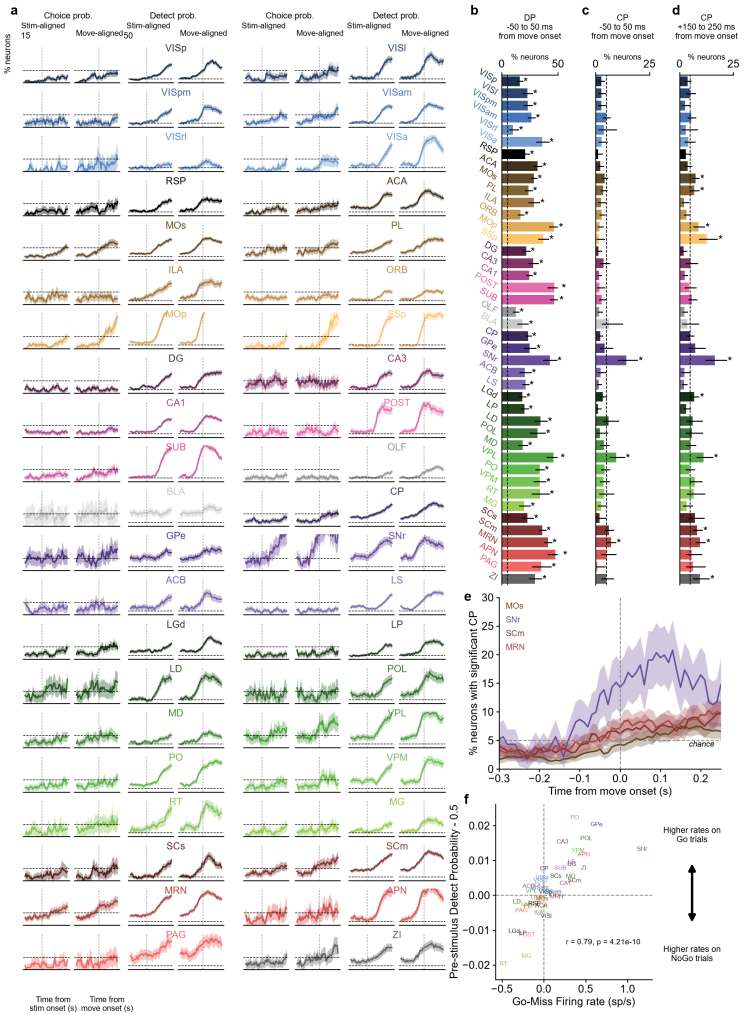
Panel	Agreement	What the comparison shows
a	Close	Same variance explained per region.
b	Close	Same order of Go, Miss and Passive before the stimulus in nearly all regions.

Extended Data Figure 7

Original



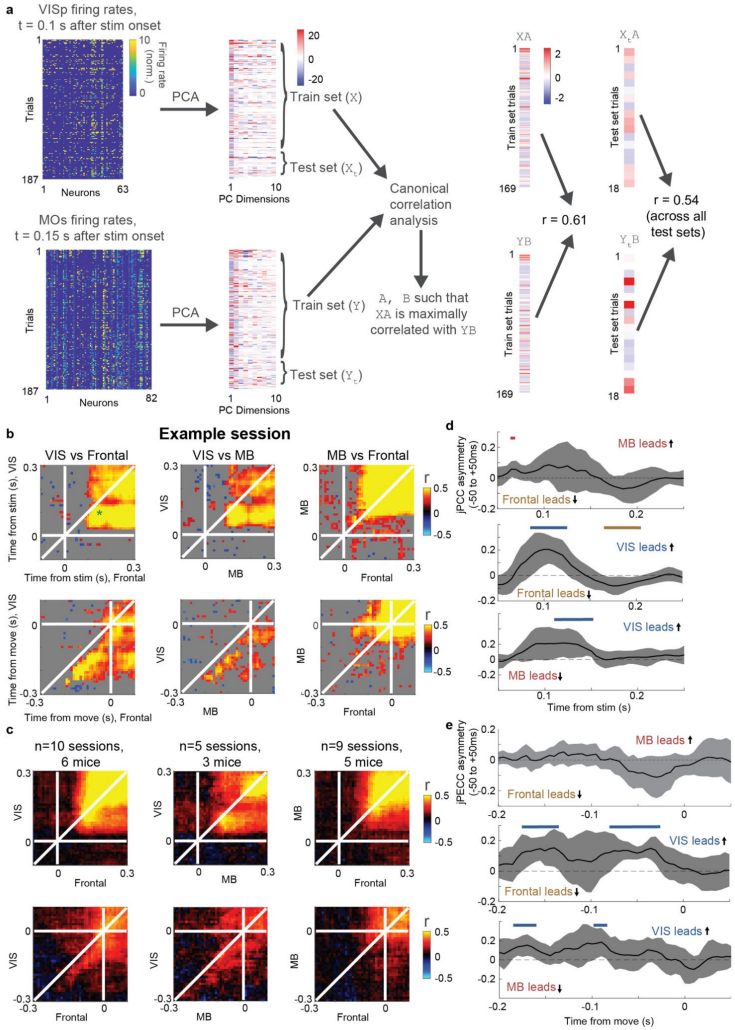
Reproduced from the full 39-session run



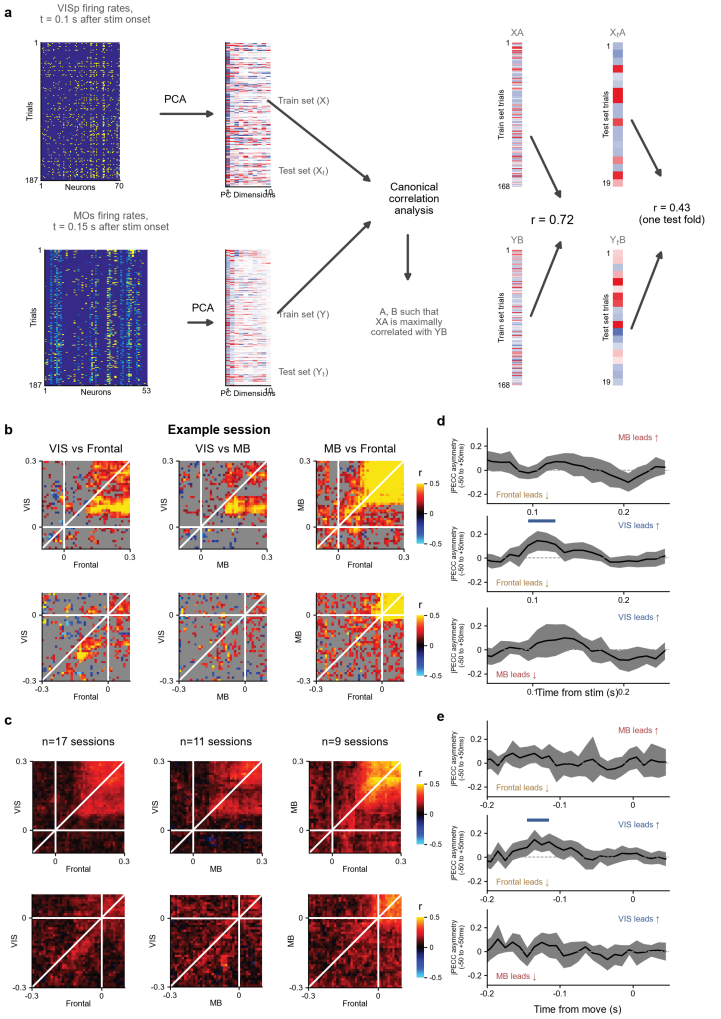
Panel	Agreement	What the comparison shows
a	Close	CP stays near chance in most regions, DP rises with movement.
b	Close	Same significant regions, broadly distributed.
c	Close	Choice probability around movement restricted to a similar set.
d	Close	Late choice signal more widespread.
e	Close	SNr reaches about 20%, others about 10%, as in the paper.
f	Close	r = 0.79 against 0.69 across all 39 sessions, and 0.81 on the 33 CP sessions.

Extended Data Figure 8

Original



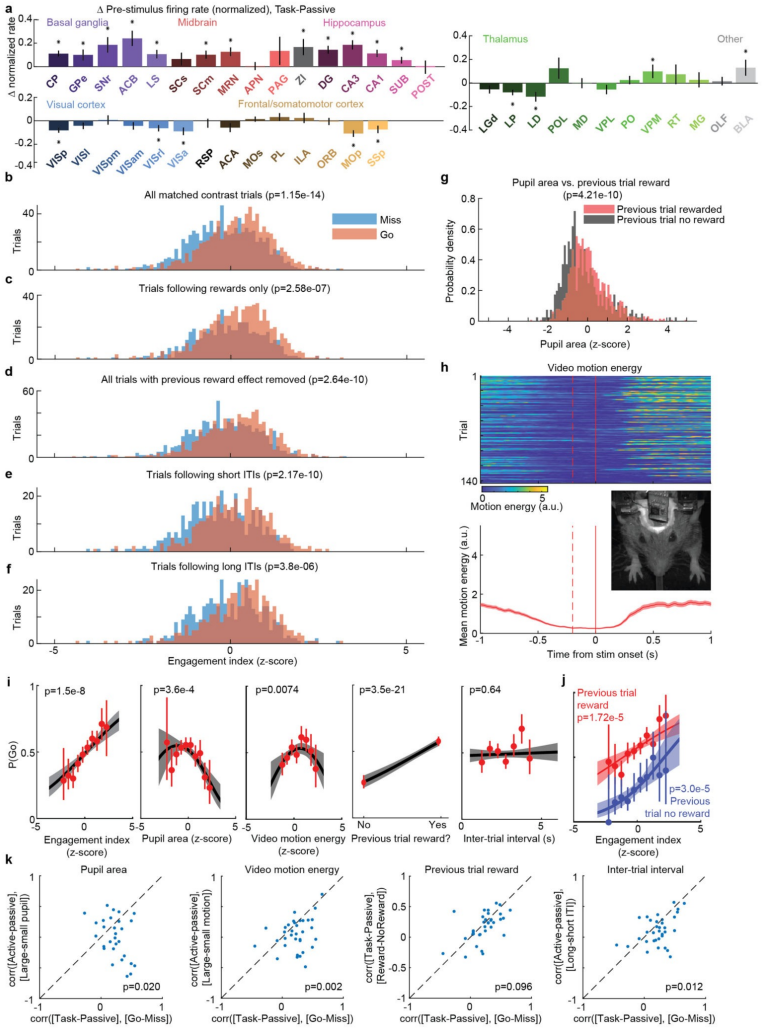
Reproduced from the full 39-session run



Panel	Agreement	What the comparison shows
a	Close	Worked example from Radnitz_2017-01-08, which has the paper's 187 trials. r = 0.72 train and 0.43 test, against 0.61 and 0.54.
b	Partial	Same session type and color scale, noisier than the paper's example.
c	Close	Same off-diagonal bias. n = 17, 11, 9 against 10, 5, 9.
d	Close	VIS leads near 0.1 s, as in the paper, with fewer significant bins.
e	Partial	VIS leads before movement, but significant in a much shorter stretch.

Extended Data Figure 9

Original



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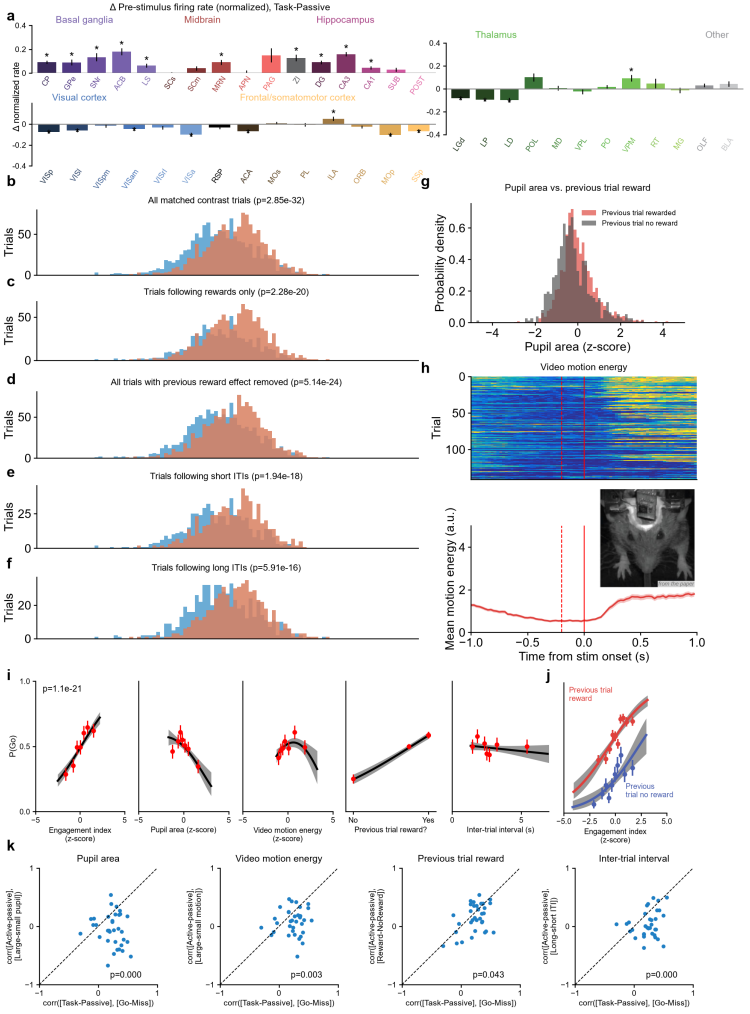

Panel	Agreement	What the comparison shows
a	Close	22 regions pass $p < 0.0012$. Same signs by group.
b-f	Close	Go above Miss in every split. Our p-values are stronger than the paper's.
g	Close	Pupil larger after reward.
h	Close	Real motion energy from a 141-trial session; the photo is from the paper.
i	Partial	Same shapes. The engagement test is far stronger here, p = 1e-21 against 1.5e-8.
j	Close	Same separation by previous reward.
k	Close	Go-Miss vector closer to engagement. Some p-values differ.

8 Reproducing this report

Figures and this report rebuild from saved results, with no new analysis. From the `Task` folder:

```
env/Scripts/python.exe analysis/run_all.py
env/Scripts/python.exe report/make_report.py
```

The analysis itself runs only with `run_all.py --analyze`.

References

[1] Distributed coding of choice, action and engagement across the mouse brain. *Nature* 576, 266–273 (2019). doi:10.1038/s41586-019-1787-x.

[2] Dataset for [1]. figshare, doi:10.6084/m9.figshare.9598406.

[3] Analysis code for [1]. `https://github.com/nsteinme/steinmetz-et-al-2019`.

[4] Allen Mouse Brain Common Coordinate Framework, version 3 (2017).

[5] Glmnet for Matlab, version used in [1].